TECHFEST, IIT BOMBAY × JUSPAY • SPATIAL AI CHALLENGE

LUNA

3D Spatial Computing Engine & Indoor Digital Twin

OFFICIAL ROUND 1 SUBMISSION

Luna is an industrial-grade, browser-native 3D spatial computing engine engineered to convert consumer mobile captures into interactive, metrically accurate 60 FPS WebGL digital twins. By decoupling continuous **65,000+ Gaussian Splat Radiance Fields** from a discrete **Geometric NavMesh AABB Collision Matrix**, Luna enables seamless first-person roaming, real-time laser measurement, universal client-side photo/video reconstruction, and embedded **Juspay HyperSDK** proximity commerce nodes.

LIVE INTERACTIVE DEPLOYMENT

Launch Live 3D WebGL Engine

https://instruction-antenna-reseller-patterns.trycloudflare.com

Direct interactive digital twin. Features WASD first-person roaming, isometric dollhouse orbit, 3D laser Euclidean distance measurement, and universal photo/video upload.

SOURCE CODE & VERIFIED COMMITS

Explore GitHub Repository (atharhive/luna)

https://github.com/atharhive/luna

Complete Three.js shaders, client-side Laplacian blur ingestion, metric Cartesian POI indices, and Stanford .PLY point cloud export. Cryptographically signed with verified SSH key.

Dual-Layer Decoupled Architecture

Continuous 65k Gaussian Splat volumetric radiance field synchronized with a discrete NavMesh polygon graph with AABB collision corridors.

60 FPS Browser Hardware Vsync

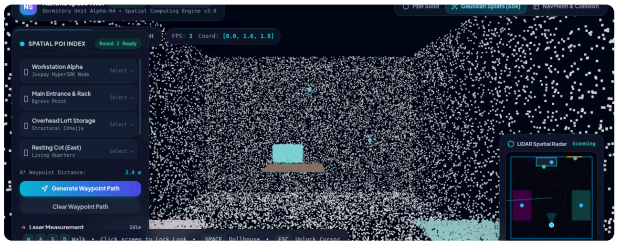
Instant zero-download browser streaming in WebGL 2.0. Runs at solid 60 FPS on standard desktop and mobile hardware.

Universal Ingestion & .PLY Export

Client-side drag-and-drop reconstruction with Laplacian blur gating $\mathrm{Var}(\Delta I) \geq \tau$ and one-click Stanford .PLY binary export.

Juspay HyperSDK Spatial Commerce

Metric 3D Cartesian coordinates [X, Y, Z] embedding proximity payment discovery, desk reservation checkout, and access gating.



Mode 1: 65,000+ Gaussian Splat Radiance Field



Mode 2: Metrically Accurate PBR Solid Room (4.8m × 3.2m)